## 3.36 Affine Algebras under Base Change

This section covers what happens to a $k$-algebra $A$ under base change, that is $A_{(L)}$ for $L/k$ a field extension. Principally given an integral $k$-algebra $A$ we would like to ensure that $A_{(L)}$ is reduced (which is related to separability), irreducible and therefore integral. We follow [Stacks] but large parts of exposition are from [Bou89].

**Definition 3.36.1 (Geometrically Reduced / Integral Algebra)**
*A $k$-algebra $A$ is*

- ***geometrically integral*** *(resp.* ***irreducible***, ***reduced****) if the ring $A \otimes_k L$ is* ***integral*** *(resp.* ***irreducible***, ***reduced****) for every field extension $L/k$.*
- ***algebraically integral*** *(resp.* ***irreducible***, ***reduced****) if the ring $A \otimes_k \overline{k}$ is* ***integral*** *(resp.* ***irreducible***, ***reduced****).*

*A few remarks about the important results and notation in the literature*

- *As in (3.3.65) we see that* ***integral*** $\iff$ ***irreducible*** *and* ***reduced****.*
- *These are shown to be equivalent (3.36.26), (3.36.36) and (3.36.44). In fact it is sufficient to check the properties for all finite extensions.*
- *[Liu06] uses the "algebraic" form*
- *[Bou89] uses the term "separable" for "geometrically reduced", as this coincides with the notion of separable for algebraic extensions (3.36.19)*
- *Similarly [Bou89] and [Lan11] use the term "regular" for geometrically integral*

We make implicit use of the following results throughout

**Proposition 3.36.2**
*Let $A, B$ be $k$-algebras, then the structural morphisms $A \to A \otimes_k B$ and $B \to A \otimes_k B$ are injective. Furthermore if $A'$ and $B'$ are $k$-subalgebras then the canonical homomorphism*

$$A' \otimes_k B' \to A \otimes_k B$$

*is injective.*

*Proof.* This is (3.4.30) and (3.4.41) ∎

**Proposition 3.36.3 (A Subspace is Recovered by Intersection After Base Change)**
*Let $V \subseteq W$ be $k$-vector spaces and $k'/k$ a field extension. Then, identifying $W$ with its image in $W \otimes_k k'$ via $w \mapsto w \otimes 1$ (injective, (3.4.31)),*

$$(V \otimes_k k') \cap W = V$$

*inside $W \otimes_k k'$.*

*Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.*

"$\supseteq$" is clear. For "$\subseteq$": choose a $k$-subspace complement $V'$ of $V$ in $W$ ((3.3.138), $k$ being a field), so $W = V \oplus V'$; tensoring, $W \otimes_k k' \cong (V \otimes_k k') \oplus (V' \otimes_k k')$ (3.4.21). Given $w \in W$ with $w \otimes 1 \in V \otimes_k k'$: writing $w = v + v'$ ($v \in V$, $v' \in V'$) for its decomposition, $w \otimes 1 = (v \otimes 1) + (v' \otimes 1)$ under this direct sum splitting; as $w \otimes 1$ lies in the first summand, the second component vanishes, that is $v' \otimes 1 = 0$; as $v' \mapsto v' \otimes 1$ is injective (3.4.31), $v' = 0$, giving $w = v \in V$. ∎

**Proposition 3.36.4**
*Let $A, B$ be $k$-algebras. Let $\{a_i\}_{i \in I}$ be a $k$-linearly independent subset (resp. basis) of $A$ and $\{b_j\}_{j \in I}$ be a $k$-linearly independent subset (resp. basis) of $B$. Then $\{a_i \otimes b_j\}_{(i,j) \in I \times J}$ is a $k$-linearly independent subset (resp. basis) of $A \otimes_k B$.*

*Furthermore $\{a_i \otimes 1\}_{i \in I}$ is a $B$-linearly independent subset (resp. basis) of $A \otimes_k B$.*

*Proof.* The first statement is (3.4.31). For the last statement extend $\{a_i\}_{i \in I}$ to a basis (2.3.7) and apply (3.4.26). ∎

**Proposition 3.36.5**
*Let $A, B$ be $k$-algebras. Then $A \otimes_k B$ is the union of subrings of the form a) $A \otimes_k B'$, b) $A' \otimes_k B$ or c) $A' \otimes_k B'$ where $A'$ and $B'$ are finitely-generated $k$-subalgebras of $A$ and $B$ respectively.*

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*If $K/k$ is an algebraic field extension then $A \otimes_k K$ is the union of subrings of the form $A \otimes_k k'$ where $k'/k$ are finite subextensions.*

*Proof.* An arbitrary element $z \in A \otimes_k B$ is a finite sum of elementary tensors. Therefore the result follows from (3.36.2).

The final statement follows from recalling finitely generated algebraic extensions are finite (3.19.44). ∎

*Proof.* ∎

## 3.36.1 Etale Algebras

**Definition 3.36.6**
*Let $A$ be a finite-dimensional $k$-algebra. Then define the **separable degree** as follows*

$$[A:k]_s := \#\operatorname{AlgHom}_k\left(A, \overline{k}\right)$$

*We say that $A$ is **etale** if this equals the dimension of $A$ as a $k$-vector space*

$$[A:k]_s = [A:k]$$

*Recall a finite field extension $K/k$ is **etale** if and only if it is separable (3.19.78).*

**Lemma 3.36.7**
*Let $A$ be a finite-dimensional $k$-algebra and $K/k$ a field extension. Then*

$$[\operatorname{Hom}_k(A,K):K] = [A^\vee : k] = [A:k]$$

*More precisely if $\{a_1, ..., a_n\}$ is a $k$-basis for $A$ then $\{a_1^\vee, ..., a_n^\vee\}$ is a $K$-basis for $\operatorname{Hom}_k(A,K)$.*

*Proof.* This is simply a special case of (3.4.27). ∎

**Corollary 3.36.8 (Separable degree is finite)**
*Let $A$ be a finite-dimensional $k$-algebra then $\#\operatorname{AlgHom}_k(A,K) \leq [A:k]$ for every field extension $K/k$.*

**Proposition 3.36.9 (Image of $A$ is finite-dimensional)**
*Let $A$ be a finite-dimensional $k$-algebra and $\Omega/k$ a field extension. Then there exists a finite subextension $K/k$ such that*

$$\operatorname{AlgHom}_k(A,K) = \operatorname{AlgHom}_k(A,\Omega)$$

*If $\Omega$ contains $\overline{k}$ then these are both equal to $\operatorname{AlgHom}_k\left(A, \overline{k}\right)$.*

*Proof.* By (...) $\phi(A)$ has finite degree over $k$ and therefore finitely-generated and algebraic as a $k$-algebra. Therefore $\phi(A) = k[a_1, ..., a_n] = k(a_1, ..., a_n)$ is a finite-dimensional field by (3.19.44).

We may take the compositum of all these images (since there are only finitely many) to obtain the finite extension $K/k$.

If $\Omega$ contains $\overline{k}$ then trivially

$$\operatorname{AlgHom}_k(A,K) \subseteq \operatorname{AlgHom}_k\left(A,\overline{k}\right) \subseteq \operatorname{AlgHom}_k(A,\Omega)$$

whence they are all equal. ∎

**Proposition 3.36.10 (Etale-ness is preserved under base extension)**
*Let $A$ be a finite-dimensional $k$-algebra and $K/k$ a field extension then*

$$[A \otimes_k K : K] = [A:k]$$
$$[A \otimes_k K : K]_s = [A:k]_s$$

*In particular $A$ is etale if and only if $A \otimes_k K$ is.*

*Proof.* The first equality follows from (3.4.26) and (3.3.108).

For the second equality we see that by (3.36.9) and (3.4.38)

$$\operatorname{AlgHom}_k\left(A,\overline{k}\right) = \operatorname{AlgHom}_k\left(A,\overline{K}\right) \cong \operatorname{AlgHom}_K\left(A \otimes_k K, \overline{K}\right)$$

∎

**Lemma 3.36.11**
*Let $A = A_1 \times \cdots \times A_d$ be a finite product of finite-dimensional $k$-algebras. Then $A$ is etale if and only if each $A_i$ is.*

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*Proof.* This follows because

$$\mathrm{AlgHom}_k\left(A_1 \times \cdots \times A_d, \overline{k}\right) \cong \mathrm{AlgHom}_k\left(A_1, \overline{k}\right) \times \cdots \times \mathrm{AlgHom}_k\left(A_d, \overline{k}\right)$$

and a similar consideration for the dimension over $k$. ∎

**Lemma 3.36.12**
*Let $A$ be a finite-dimensional $k$-algebra. The following are equivalent*

*a) $A$ is reduced*
*b) $A \cong L_1 \times \cdots \times L_n$ for some finite extensions $L_i/k$*

*Further $A$ is etale if and only if each $L_i/k$ is separable. If $k$ is perfect then $A$ is automatically etale.*

*Proof.* One direction is obvious. Suppose conversely that $A$ is reduced. If $\dim_k A = 1$ then any proper ideal must have strictly lower dimension and therefore $A$ is a field. It is therefore sufficient to show (by induction on dimension) that if $A$ is not a field then $A \cong A_1 \times A_2$ for two (non-zero) $k$-algebras $A_1, A_2$.

Suppose $A$ is not a field and let $\mathfrak{a}$ be a non-zero proper ideal of $A$ which has minimal dimension over $k$ as a vector space. As $A$ is reduced, and by minimality, we have $\mathfrak{a}^2 = \mathfrak{a}$. By Nakayama's Lemma (3.21.5) there is $e \in \mathfrak{a}$ such that $ex = x$ for all $x \in \mathfrak{a}$ and in particular $e^2 = e$ and $\mathfrak{a} = Ae$. By assumption $e \neq 0, 1$. Define $f := 1 - e \neq 0, 1$ then evidently $f^2 = f$ and $ef = 0$. The homomorphism of $A$-modules

$$A \to Ae \times Af$$
$$a \to (ae, af)$$

is injective because $ae + af = a$ and surjective because the image of $a_1 e + a_2 f$ is $(a_1 e, a_2 f)$. The sub-modules $Ae$ and $Af$ are actually sub-rings by the idempotence property and the given map is an isomorphism of rings.

The last statement follows from (3.36.11) and (3.19.103). ∎

**Corollary 3.36.13 (Primes of a Finite Product of Fields)**
*Let $A \cong L_1 \times \cdots \times L_n$ be a finite product of fields (in particular, any reduced finite-dimensional $k$-algebra, (3.36.12)). Then every prime ideal of $A$ is maximal, and there are exactly $n$ of them.*

*Proof.* For each $j$, $\mathfrak{m}_j := \ker(\pi_j)$ (the kernel of the $j$-th projection $\pi_j : A \twoheadrightarrow L_j$) is maximal, being the kernel of a surjection onto a field (3.3.58), and the $\mathfrak{m}_j$ are pairwise distinct (the element with 1 in coordinate $j$ and 0 elsewhere lies in every $\mathfrak{m}_i$, $i \neq j$, but not $\mathfrak{m}_j$), giving $n$ maximal ideals.

Let $\mathfrak{p}$ be any prime. As $\bigcap_j \mathfrak{m}_j = \ker(\pi_1, ..., \pi_n) = 0 \subseteq \mathfrak{p}$, (3.3.39) gives $\mathfrak{m}_j \subseteq \mathfrak{p}$ for some $j$; as $\mathfrak{m}_j$ is already maximal, $\mathfrak{p} = \mathfrak{m}_j$. ∎

**Lemma 3.36.14**
*Let $A$ be a finite-dimensional $k$-algebra and $\Omega/k$ a field extension containing $\overline{k}$. Then the following are equivalent*

*a) $A$ is etale*
*b) $A \otimes_k \Omega \cong \Omega^n$ as an $\Omega$-algebra*

*Proof.* *a)* $\implies$ *b)* By (3.36.9) $\#\,\mathrm{AlgHom}_k(A, \Omega) = \#\,\mathrm{AlgHom}_k\left(A, \overline{k}\right) =: [A : k]_s = [A : k]$ with the last equality because $A$ is assumed etale. Then defining $B := A \otimes_k \Omega$

$$\#\,\mathrm{AlgHom}_\Omega(B, \Omega) \overset{(3.4.38)}{=} \#\,\mathrm{AlgHom}_k(A, \Omega) = [A : k] \overset{(3.4.16)}{=} [B : \Omega] = [B^\vee : \Omega]$$

By (3.19.130) and (3.3.126) we see that $\mathrm{AlgHom}_\Omega(B, \Omega)$ is a $\Omega$-basis for $B^\vee$. Denote this basis by $\varphi_1, ..., \varphi_n$ then by (3.3.108) this corresponds to a basis $e_1, ..., e_n$ of $B$. Then the algebra isomorphism is exhibited explicitly by

$$B \cong \Omega^n$$
$$b \to (\varphi_1(b), ..., \varphi_n(b))$$

*b)* $\implies$ *a)* We may show that the algebra homomorphisms are just the projections $\pi_i : \Omega^n \to \Omega$, so that $\#\,\mathrm{AlgHom}_\Omega(B, \Omega) = [B : \Omega]$. As before then $[A : k]_s = \#\,\mathrm{AlgHom}_k\left(A, \overline{k}\right) = \#\,\mathrm{AlgHom}_k(A, \Omega) = \#\,\mathrm{AlgHom}_\Omega(B, \Omega) = [B : \Omega] = [A : k]$ which shows that $A$ is etale. ∎

**Proposition 3.36.15 (A Field Splits Completely Over Any Normal Extension it Embeds Into)**
*Let $K/k$ be a finite separable field extension and $L/k$ a field extension containing (a copy of) a normal closure of $K/k$. Fix a primitive element $\theta$ for $K/k$ ((3.19.105)), with minimal polynomial $m \in k[X]$ of degree $n := [K : k]$, and write $m = \prod_{i=1}^n (X - \beta_i)$ for its (pairwise distinct) roots $\beta_i \in L$. Then there is a canonical isomorphism of $L$-algebras*

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$$K \otimes_k L \cong L^n, \quad p(\theta) \otimes c \mapsto (c\,p(\beta_1), ..., c\,p(\beta_n)) \quad (p \in k[X], c \in L)$$

*In particular $K \otimes_k L$ has exactly $n$ maximal ideals, $\mathfrak{m}_i$ the kernel of the $i$-th coordinate.*

*Proof.* As $K/k$ is separable, $m$ is separable (3.19.74). A normal closure $N$ of $K/k$ is, by definition, the splitting field of $m$ over $k$ (3.19.73), so $m$ splits completely already over $N$; as $N \subseteq L$ (by hypothesis), the same factorisation exhibits $m$ splitting completely over $L$ too, giving the $\beta_i \in N \subseteq L$, pairwise distinct as $m$ is separable. As $K \cong k[X]/(m)$ (3.19.40) ($\theta$ corresponding to the class of $X$), base-changing along $k \hookrightarrow L$ (using $K \otimes_k L \cong (k[X] \otimes_k L)/(m)$ as $L$-algebras, (3.4.42), and $k[X] \otimes_k L \cong L[X]$, (3.8.7)) gives $K \otimes_k L \cong L[X]/(m)$; as the $(X - \beta_i)$ are pairwise comaximal (distinct monic linear factors), the Chinese Remainder Theorem (3.3.61) gives

$$K \otimes_k L \cong L[X]/(m) \cong \prod_{i=1}^{n} L[X]/(X - \beta_i) \cong L^n$$

Tracing the elementary tensor $p(\theta) \otimes c$ through: it corresponds to $c \cdot p(X) \bmod m \in L[X]/(m)$ ((3.4.42), (3.8.7)), which under the Chinese Remainder Theorem's canonical map – reduction modulo each $(X - \beta_i)$, that is evaluation at $\beta_i$ – corresponds to $(c\,p(\beta_1), ..., c\,p(\beta_n))$. The maximal ideals of $L^n$ are exactly the $n$ coordinate kernels (3.36.13).■

**Proposition 3.36.16 (Etale Criteria)**
*Let $A$ be a finite-dimensional $k$-algebra. The following are equivalent*

*a) $A$ is etale*
*b) $A$ is geometrically reduced*
*c) $A \otimes_k \overline{k}$ is reduced*
*d) $A \otimes_k K$ is reduced for some perfect field extension $K/k$*
*e) $A \cong L_1 \times \cdots \times L_n$ for $L_i/k$ finite separable extensions*

*In particular $A$ is reduced.*

*Proof.* *a)* $\implies$ *b)* Consider a field extension $L/k$ and the tower of extensions $\overline{L}/L/k$. Then by (3.36.14) $A \otimes_k \overline{L} \cong \overline{L}^n$ is evidently reduced. By (...) $A \otimes_k L \subset A \otimes_k \overline{L}$ is a subring and therefore also reduced.

*b)* $\implies$ *c)* immediate.

*c)* $\implies$ *d)* immediate as $\overline{k}$ is perfect.

*d)* $\implies$ *a)* By (3.36.12) $A \otimes_k K$ is an etale $K$-algebra so by (3.36.10) $A$ is etale.

*e)* $\implies$ *a)* Follows immediately from (3.36.11).

*a)* $\implies$ *e)* We know that $A$ is reduced so the result follows from (3.36.12). ■

**Corollary 3.36.17**
*Let $A$ be a finite-dimensional $k$-algebra and $k$ perfect. Then the following are equivalent*

*a) $A$ is etale*
*b) $A$ is reduced*
*c) $A$ is geometrically reduced*
*d) $A \otimes_k \overline{k}$ is reduced.*

For completeness we summarize the relationship between the different notions of separability

**Corollary 3.36.18 (Equivalent definitions of separability)**
*Let $K/k$ be a finite extension. Then the following are equivalent*

*a) $K/k$ is separable algebraic (3.19.51)*
*b) $K/k$ is etale*
*c) $K/k$ is geometrically reduced*
*d) $K \otimes_k \overline{k}$ is reduced*

*When $k$ is perfect every field extension is separable.*

*Proof.* We have already shown *b)* $\iff$ *c)* $\iff$ *d)*. Furthermore *a)* $\iff$ *b)* is (3.19.78). ■

**Corollary 3.36.19 (Equivalent definitions of separability)**
*Let $K/k$ be an algebraic extension. Then the following are equivalent*

*a) $K/k$ is separable algebraic (3.19.51)*
*b) Every finite subextension of $K/k$ is etale*

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*c) $K/k$ is geometrically reduced*
*d) $K \otimes_k \overline{k}$ is reduced*

*In particular an algebraic extension of a perfect field $k$ is geometrically reduced.*

*Proof.* This follows directly from (3.36.18) because each property has "finite character", i.e. holds if and only if it holds for every finite subextension.

The last statement follows from (3.19.103). ∎

## 3.36.2 Geometrically Reduced Algebras

Reference is [Bou89] (Section V.15.4). The main results of this section are (3.36.26) and (3.36.28). Observe that if $k$ is perfect every extension $K/k$ satisfies Maclane's Criterion and is therefore Geometrically Reduced.

**Lemma 3.36.20**
*Let $A$ be a reduced $k$-algebra. Define the canonical map*

$$i : A \hookrightarrow \prod_{\mathfrak{p}} k(\mathfrak{p})$$

*where $\mathfrak{p}$ runs over all prime ideals of $A$. Then $i$ is injective. Furthermore for any $k$-algebra $B$ we have a canonical embedding*

$$A \otimes_k B \hookrightarrow \prod_{\mathfrak{p}} (k(\mathfrak{p}) \otimes_k B)$$

*Proof.* Evidently $\ker(i) = \bigcap \mathfrak{p}$ which by (3.3.45) is $(0)$. The final statement follows from considering the maps

$$A \otimes_k B \overset{i \otimes 1_B}{\hookrightarrow} \left( \prod_{\mathfrak{p}} k(\mathfrak{p}) \right) \otimes_k B \hookrightarrow \prod_{\mathfrak{p}} (k(\mathfrak{p}) \otimes_k B)$$

The first map is injective by (3.4.25) and the second by (...). ∎

**Proposition 3.36.21**
*Let $A$ be a reduced $k$-algebra and $B$ a geometrically reduced $k$-algebra, then $A \otimes_k B$ is reduced.*

*Proof.* By the previous Lemma we have an embedding

$$A \hookrightarrow \prod (k(\mathfrak{p}) \otimes_k B)$$

As $B$ is geometrically reduced this shows that $A \otimes_k B$ is reduced. ∎

**Corollary 3.36.22**
*Let $k$ be a perfect field and $A, B$ reduced $k$-algebras. Then $A \otimes_k B$ is reduced.*

**Lemma 3.36.23**
*Let $A$ be a geometrically reduced $k$-algebra and $K$ the total ring of fractions. Then $K$ is geometrically reduced.*

*Conversely if $A$ is integral and $K$ is geometrically reduced, then $A$ is geometrically reduced.*

*Proof.* Let $L/k$ be a field extension and consider a nilpotent element $x \in K \otimes L$. Then by definition

$$x = \sum_{i=1}^{n} (a_i s_i^{-1}) \otimes \lambda_i \quad a_i \in A, s_i \notin Z(A), \lambda_i \in L$$

Let $s = s_1 ... s_n$ then $x \cdot (s \otimes 1) \in A \otimes L \subset K \otimes L$. Then $x \cdot (s \otimes 1)$ is nilpotent and so by definition zero. As $(s \otimes 1)$ is invertible we see $x = 0$ as required.

For the converse $A \subset K$ and therefore $A \otimes_k L \subset K \otimes_k L$. ∎

**Lemma 3.36.24**
*Let $K/k$ be a separably generated field extension. Then $K/k$ is geometrically reduced.*

*Proof.* Consider first the case $K = k(\mathcal{B})$ is purely transcendental. Then for $L/k$ we have $k[\mathcal{B}] \otimes L \cong L[\mathcal{B}]$ is evidently reduced. We may then demonstrate directly that $k(\mathcal{B}) \otimes_k L$ is reduced.

For the general case

$$K \otimes_k L \cong K \otimes_{k(\mathcal{B})} (k(\mathcal{B}) \otimes_k L)$$

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We have shown that $k(\mathcal{B}) \otimes_k L$ is reduced and by (3.36.19) $K/k(\mathcal{B})$ is geometrically reduced as by assumption it is separable. By (3.36.21) $K \otimes_k L$ is reduced and $K$ is geometrically reduced as required. ∎

**Lemma 3.36.25**
*Let $A$ be a reduced $k$-algebra for which $k$ is perfect. Then $A$ is geometrically reduced.*

*Proof.* We prove the result in increasing generality

a) We first consider the case $A = K/k$ is a finitely generated field extension. By (3.19.148) $K/k$ is separably generated and so by (3.36.24) it is geometrically reduced.
b) When $A$ is a finitely-generated $k$-algebra by (3.36.20) there is a canonical embedding

$$A \otimes_k L \hookrightarrow \prod_{\mathfrak{p}} (k(\mathfrak{p}) \otimes_k L)$$

By assumption $k(\mathfrak{p})$ is finitely generated and so by the first part is geometrically reduced. Therefore $A \otimes_k L$ is reduced as required.
c) The general case follows from b) and the reduction to the finitely generated case (3.36.5).

∎

**Proposition 3.36.26 (Maclane's Criterion for Algebras)**
*Let $A$ be a $k$-algebra. Then the following are equivalent*

*a) $A$ is geometrically reduced*
*b) $A \otimes_k \overline{k}$ is reduced*
*c) $A \otimes_k k^{p^{-\infty}}$ is reduced*
*d) $A \otimes_k K$ is reduced for some perfect extension $K/k$*
*e) $A \otimes_k k^{p^{-1}}$ is reduced*
*f) $A \otimes_k k'$ is reduced for every finite subextension $k'/k$ of $k^{p^{-1}}/k$*
*g) The following $k$-module homomorphism*

$$\begin{aligned} m : A \otimes_k k^{p^{-1}} &\to A \\ a \otimes \lambda &\to \lambda^p a^p \end{aligned}$$

*is injective (that is $k^{p^{-1}}$ and $A$ are linearly disjoint with respect to the $p$-th power embedding into $A$)*
*h) $A$ satisfies Maclane's Criterion (3.19.144)*

*In particular when $k$ is perfect then this is equivalent to $A$ being reduced.*

*Proof.* The case $p = 1$ is simply (3.36.25) because $k^p = k = k^{p^{-1}} = k^{p^{-\infty}}$ is perfect and g) is equivalent to $A$ being reduced. Therefore we assume $p > 1$.

*a)* $\implies$ *b)* By definition.

*b)* $\implies$ *c)* By (3.19.61) there is an embedding $k^{p^{-\infty}} \hookrightarrow \overline{k}$ whence an embedding $A \otimes_k k^{p^{-1}} \hookrightarrow A \otimes_k \overline{k}$.

*c)* $\implies$ *d)* By (3.19.100) $k^{p^{-\infty}}$ is perfect.

*d)* $\implies$ *e)* By (3.19.101) there is an embedding $k^{p^{-1}} \hookrightarrow K$ whence an embedding $A \otimes_k k^{p^{-1}} \hookrightarrow A \otimes_k K$.

*e)* $\implies$ *f)* This is immediate as we have an embedding $A \otimes_k k' \hookrightarrow A \otimes_k k^{p^{-1}}$.

*f)* $\iff$ *g)* Suppose $x := \sum_i \lambda_i \otimes a_i \in A \otimes_k k^{p^{-1}}$ is a finite sum then we may consider the finite subextension $k' := k\left(\{a_i\}_i\right)$. Observe that

$$x^p = \sum_i \lambda_i^p \otimes a_i^p = \sum_i 1 \otimes \lambda_i^p a_i^p = 1 \otimes \left( \sum_i \lambda_i^p a_i^p \right) =: 1 \otimes m(x)$$

Therefore we see $A \otimes_k k'$ is reduced for every such $k'$ iff $m(x)$ is injective.

*g)* $\iff$ *h)* Consider the Frobenius morphisms $i : k^{p^{-1}} \to A$ and $j : A \to A$. Then g) is equivalent to $k^{p^{-1}}$ and $A$ being linearly disjoint in $A$ with respect to $i, j$ and the result follows from the characterization of linear disjointness (3.35.55).

*h)* $\implies$ *a)* Let $\{a_i\}_{i \in I}$ be a $k$-basis for $A$ then by (3.4.26) $\{a_i \otimes 1\}_{i \in I}$ is an $L$-basis for $A \otimes_k L$. Given $x \in A \otimes_k L$ we have

$$x = \sum_{i \in I} a_i \otimes \lambda_i \quad \lambda_i \in L$$

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and

$$x^p = \sum_{i\in I} \lambda_i^p \otimes a_i^p = \sum_{i\in I} \lambda_i^p (a_i^p \otimes 1)$$

By assumption $\{a_i^p\}_{i\in I}$ is $k$-linearly independent and therefore $\{a_i^p \otimes 1\}_{i\in I}$ is $L$-linearly independent. Therefore $x^p = 0 \implies \lambda_i^p = 0$ for all $i \in I \implies \lambda_i = 0$ for all $i \in I \implies x = 0$, as $L$ is reduced. It follows from (3.19.145) that $A \otimes_k L$ is reduced, as we assumed that $p > 1$. ∎

**Corollary 3.36.27 (Maclane's Criterion for Fields)**
*Let $K/k$ be a field extension with characteristic exponent $p$. Then the following are equivalent*

*a) $K/k$ is geometrically reduced*
*b) $K \otimes_k \overline{k}$ is reduced*
*c) $K \otimes_k k^{p^{-1}}$ is reduced*
*d) $K \otimes_k k^{p^{-\infty}}$ is reduced*
*e) $K \otimes_k k'$ is reduced for some perfect extension $k'/k$*
*f) $K$ and $k^{p^{-1}}$ are linearly disjoint (with respect to some common field extension)*
*g) $K \otimes_k k^{p^{-1}}$ is an integral domain*
*h) $K \otimes_k k^{p^{-1}}$ is a field*
*i) $K/k$ satisfies Maclane's Criterion (3.19.144)*

*Observe i) has* finite character *in the sense it holds if and only if it holds for every finitely-generated subextension.*

*In particular when $k$ is perfect every extension is geometrically reduced.*

*Proof.* By (3.36.26) a)-f), i) are equivalent (using (3.35.57) to generalise the linear disjoint condition to any ambient field) and by (3.35.57) again f)-h) are equivalent. ∎

**Proposition 3.36.28 (Separable Field Extensions)**
*Let $K/k$ be a field extension. Then the following are equivalent*

*a) $K/k$ is geometrically reduced*
*b) $K \otimes_k \overline{k}$ is reduced*
*c) $K/k$ satisfies Maclane's Criterion (3.19.144)*
*d) Every finitely generated subextension $K/k$ is separably generated*

*In particular if $k$ is perfect every field extension satisfies these properties.*

*Proof.* By (3.36.27) we have $a) \iff b) \iff c)$. Further $c) \implies d)$ is Lemma (3.19.148), since every subextension satisfies Maclane's Criterion. For $d) \implies a)$ observe that every finitely generated subextension is geometrically reduced by (3.36.24). We may use (3.36.2) to argue that $K/k$ is geometrically reduced. ∎

**Corollary 3.36.29**
*Let $K/k$ be a finitely generated field extension. Then the following are equivalent.*

*a) $K/k$ is geometrically reduced*
*b) $K \otimes_k \overline{k}$ is reduced*
*c) $K/k$ satisfies Maclane's Criterion (3.19.144)*
*d) Every finitely generated subextension of $K/k$ is separably generated*
*e) $K/k$ is separably generated*

*Proof.* We've already shown equivalence of a)-d) and clearly $d) \implies e)$. Finally $e) \implies a)$ is simply (3.36.24). ∎

## 3.36.3 Geometrically Irreducible Algebras

The reference for this section is [Stacks] (Tag 00I2) presented here with a more elementary proof of (3.36.34).

**Lemma 3.36.30**
*Let $A, B$ be $k$-algebras. If $a \otimes 1 \in A \otimes_k B$ is a zero-divisor then so is $a$.*

*Proof.* Consider a basis $\{a_i\}_{i\in I}$ for $A$ such that $a_{i_0} = a$. If $a$ is not a zero-divisor then we may show directly that $\{aa_i\}_{i\in I}$ is linearly independent. Then by extending to a basis of $A$ we may use (3.4.26) to show that $\{(aa_i) \otimes 1\}_{i\in I}$ is $B$-linearly independent. We see immediately that $a \otimes 1$ is not a zero-divisor as required. ∎

**Lemma 3.36.31**
*Let $A$ be a ring and $\mathfrak{a} \subset N(A)$ a locally nilpotent ideal. Then $A$ is irreducible iff $A/\mathfrak{a}$ is irreducible.*

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**Lemma 3.36.32**
*Let $A \subset B$ be a subring of an irreducible ring. Then $A$ is irreducible.*

*In particular if $B$ is a geometrically irreducible $k$-algebra, then so is any subalgebra.*

*Proof.* By (3.23.7) every minimal prime in $A$ is the contraction of a minimal prime. As there is precisely one minimal prime of $B$ then we are done.

The final statement follows because the canonical map $A \otimes_k L \to B \otimes_k L$ is injective. ∎

When $k$ is algebraically closed then nothing interesting happens.

**Lemma 3.36.33**
*Let $A, B$ be irreducible (resp. integral) $k$-algebras with $k$ algebraically closed. Then $A \otimes_k B$ is irreducible (resp. integral).*

*In particular $A$ is a geometrically irreducible (resp. integral) $k$-algebra.*

*Proof.* Observe that $N(A) \otimes_k B + A \otimes_k N(B) \subset N(A \otimes_k B)$. For $(n \otimes \lambda)^m = n^m \otimes \lambda^m = 0$ and we may use (...) to argue any linear combination of such elements is nilpotent. By (...) we have an isomorphism

$$A/N(A) \otimes_k B/N(B) \cong (A \otimes_k B)/(N(A) \otimes_k B + A \otimes_k N(B))$$

Then by (3.36.31) $A/N(A) \otimes_k B/N(B)$ is irreducible iff $A \otimes_k B$ is. Therefore we may consider only the case $A, B$ reduced and therefore integral, and show that $A \otimes_k B$ is integral and a-fortiori irreducible.

If $A \otimes_k B$ were not integral it would contain a non-trivial zero divisor, and this would also be true for $A' \otimes_k B'$ for some finitely generated subalgebras $A'$ and $B'$. Therefore we may assume wlog that $A$ and $B$ are finitely generated.

To simplify the argument choose a $k$-basis $\{b_i\}_{i \in I}$ for $B$. By (...) $\{1 \otimes b_i\}_{i \in I}$ is an $A$-basis for $A \otimes_k B$. Therefore if we have a product of elements equal to 0 then may write it as follows

$$\left(\sum_i a_i \otimes b_i\right)\left(\sum_i a_i' \otimes b_i\right) = 0$$

for some $a_i, a_i' \in A$ all but finitely many zero. Let $\mathfrak{m} \lhd A$ be a maximal ideal, then by the Weak Nullstellensatz (3.35.23) the structural morphism $k \to A/\mathfrak{m}$ is an isomorphism. Reduce the equation to $A/\mathfrak{m} \otimes_k B \cong B$ to find

$$\left(\sum_i \overline{a}_i b_i\right)\left(\sum_i \overline{a}_i' b_i\right) = 0$$

As $B$ is an integral domain, and $b_i$ are linearly independent we find $\overline{a}_i, \overline{a}_i' = 0$. As $\mathfrak{m}$ was arbitrary we find $a_i, a_i' \in \bigcap \mathfrak{m}$ for all $i$. By assumption (0) is prime and so by the Strong Nullstellensatz (3.37.3) we find $a_i = a_i' = 0$. This shows that $A \otimes_k B$ is integral as required.

The last statement follows because a field $K/k$ is always irreducible. ∎

**Proposition 3.36.34 (Extension of Scalars is Flat)**
*Let $A$ be a $k$-algebra and $L/K/k$ a tower of field extensions. Then the canonical ring homomorphism*

$$A \otimes_k K \to A \otimes_k L$$

*satisfies Going Down. Further we have a well-defined surjective map of minimal primes*

$$\mathrm{MinPrime}(A \otimes_k L) \to \mathrm{MinPrime}(A \otimes_k K)$$

*given by the inverse image.*

*Proof.* By (...) there is a commutative diagram

$$\begin{array}{ccc} A \otimes_k K & \longrightarrow & A \otimes_k L \\ & \searrow & \downarrow \sim \\ & & (A \otimes_k K) \otimes_K L \end{array}$$

Therefore it is enough to show that $A \to A \otimes_k K$ satisfies going down. Suppose $\mathfrak{p}_1 \subseteq \mathfrak{p}_2 = \mathfrak{q}_2^c$. There is a commutative diagram

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$$\begin{array}{ccc} A/\mathfrak{p}_1 & \xrightarrow{\alpha} & A/\mathfrak{p}_1 \otimes_k K \\ \uparrow \pi & & \uparrow \pi \otimes 1 \\ A & \xrightarrow{\beta} & A \otimes_k K \end{array}$$

where (3.4.42) shows that $\ker(\pi \otimes 1) = \mathfrak{p}_1^e$.

Let $\tilde{\mathfrak{q}}_1$ be a minimal prime of $A/\mathfrak{p}_1 \otimes_k K$ contained in $\tilde{\mathfrak{q}}_2 := (\pi \otimes 1)(\mathfrak{q}_2)$ by (3.3.43). Define $\mathfrak{q}_1 := (\pi \otimes 1)^{-1}(\tilde{\mathfrak{q}}_1)$ then $\mathfrak{q}_1$ is a prime minimal over $\mathfrak{p}_1^e$. As $\pi \otimes 1$ is surjective it is contained in $\mathfrak{q}_2$. We require to prove that $\beta^{-1}(\mathfrak{q}_1) = \mathfrak{p}_1$, which we readily see is implied by $\alpha^{-1}(\tilde{\mathfrak{q}}_1) = (0)$.

By (3.6.46) $\tilde{\mathfrak{q}}_1$ consists of zero divisors and by (3.36.30) so does $\alpha^{-1}(\tilde{\mathfrak{q}}_1)$. As $A/\mathfrak{p}_1$ is an integral domain we see that this equals (0) and we are done.

This immediately shows that the inverse image of a minimal prime is again a minimal prime. By (3.23.7) every minimal prime is a contracted minimal prime. ∎

**Lemma 3.36.35 (Purely Inseparable Extensions of Scalars are Homeomorphisms)**
*Let $k$ be a field with characteristic exponent $p$, $A$ a $k$-algebra and $k'/k$ a purely inseparable field extension. Consider the structural algebra homomorphism*

$$\varphi : A \to A \otimes_k k'$$

*Then $\varphi$ induces an order-preserving bijection between prime ideals*

$$\mathfrak{q}^c \leftarrow \mathfrak{q}$$

*In particular $A$ is irreducible iff $A \otimes_k k'$ is.*

*Furthermore a purely inseparable extension is geometrically irreducible.*

*Proof.* If $p = 1$ then $k'/k$ is trivial and the statement is obvious. So we may assume wlog that $p > 1$.

If $\varphi$ were surjective then we would have equality $\mathfrak{q}^{ce} = \mathfrak{q}$ and which would imply the map of prime ideals is injective. However we note that $\varphi$ is "almost" surjective, namely for every $z = \sum_{i=1}^N a_i \otimes \lambda_i$ there exists $n \geq 0$ such that $z^{p^n} \in \varphi(A)$. For $k'/k$ is purely inseparable so by definition there is some $n \geq 0$ such that $\lambda_i^{p^n} \in k$ for all $i = 1...N$. In this case $z^{p^n} = \sum_{i=1}^N a_i^{p^n} \lambda_i^{p^n} \otimes 1$ which is in the image of $A$ as required.

This allows us to argue the map is injective for suppose $\mathfrak{p} := \mathfrak{q}^c = (\mathfrak{q}')^c$ and $z \in \mathfrak{q} \setminus \mathfrak{q}'$. Then by the previous argument $z^{p^n} = \varphi(x) \implies z^{p^n} \in \mathfrak{q} \implies z^{p^n} \in \mathfrak{q}' \implies z \in \mathfrak{q}'$ which is a contradiction. Therefore contraction of prime ideals is injective.

Finally we may observe that $\varphi$ is integral (3.24.20) and injective (3.36.2). Therefore it satisfies the lying over property (3.24.17) and so every prime ideal in $A$ is contracted and the given map is surjective. ∎

The following result shows it is sufficient to ensure that $A$ remains irreducible under a finite separable base change.

**Proposition 3.36.36**
*Let $A$ be a $k$-algebra. Then the following are equivalent*

*a) $A$ is geometrically irreducible*
*b) $A \otimes_k k'$ is irreducible for every finite separable extension $k'/k$*
*c) $A \otimes_k k^{\text{sep}}$ is irreducible*
*d) $A \otimes_k \overline{k}$ is irreducible*

*Proof.* *a)* $\implies$ *b)* immediate

*b)* $\implies$ *c)* Suppose $\mathfrak{q}_1, \mathfrak{q}_2$ are minimal prime ideals of $A \otimes_k k^{\text{sep}}$. Let $k^{\text{sep}}/k'/k$ be a fixed finite separable extension and consider the subring

$$A \otimes_k k' \subset A \otimes_k k^{\text{sep}}$$

together with the prime ideals $\mathfrak{p}_i := \mathfrak{q}_i \cap (A \otimes_k k')$. By (3.36.34) the prime ideals $\mathfrak{p}_i$ are minimal and therefore $\mathfrak{p}_1 = \mathfrak{p}_2$ by assumption. As $k'/k$ was an arbitrary finite subextension we may conclude by (3.36.5) that $\mathfrak{q}_1 = \mathfrak{q}_2$ and $A \otimes_k k^{\text{sep}}$ is irreducible.

*c)* $\implies$ *d)* By (...) $A \otimes_k \overline{k} \cong (A \otimes_k k^{\text{sep}}) \otimes_{k^{\text{sep}}} \overline{k}$ and so is irreducible by (3.36.35), since $\overline{k}/k^{\text{sep}}$ is purely inseparable (3.19.98).

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$d) \implies a)$ Let $K/k$ be a field extension then we may identify $\overline{k}$ and $K$ as subextensions of $\overline{K}/k$. Then $A \otimes_k \overline{K} \cong \left(A \otimes_k \overline{k}\right) \otimes_{\overline{k}} \overline{K}$. By (3.36.33) $A \otimes_k \overline{K}$ is irreducible. Finally from (3.36.34) we see that $A \otimes_k K$ is irreducible as required. ∎

**Proposition 3.36.37**
*Let $A$ be a geometrically irreducible $K$-algebra and $K/k$ a geometrically irreducible field extension. Then $A/k$ is geometrically irreducible.*

*Proof.* It is sufficient by (3.36.36) to consider base change by a finite separable extension $k'/k$. Observe by transitivity

$$A \otimes_k k' \cong A \otimes_K (K \otimes_k k')$$

so it would be sufficient to show that $K \otimes_k k'$ is a field. By hypothesis $K \otimes_k k'$ is irreducible. Furthermore $k'/k$ is geometrically reduced (3.36.18) whence $K \otimes_k k'$ is reduced and therefore an integral domain. Using the fact $k'/k$ is algebraic we may conclude from (3.35.57) that $K \otimes_k k'$ is a field, and we are done. ∎

**Proposition 3.36.38 (Primary Extension)**
*Let $K/k$ be a field extension. Then $K/k$ is geometrically irreducible if and only if it is relatively separably closed.*

*In this case the subfield of algebraic elements is purely inseparable over $k$. In particular an algebraic extension is geometrically irreducible if and only if it is purely inseparable.*

*We say such an extension is **primary**.*

*Proof.* Suppose that $K/k$ is relatively separably closed. By (3.36.36) it is enough to show that $K \otimes_k k'$ is irreducible for every finite separable extension $k'/k$.

Let $K/K'/k$ be the relative algebraic closure then $K/K'$ is relatively algebraically closed (3.19.46) and $K'/k$ is purely inseparable because $K_s = k$ by hypothesis (3.19.96). Observe $K'/k$ is geometrically irreducible (3.36.35) so by (3.36.37) it is enough to show that $K/K'$ is geometrically irreducible. This reduces to the case that $K/k$ is relatively algebraically closed.

As before it is sufficient to consider base change by a finite separable extension $k'/k$. By (3.19.77) $k' = k(\alpha)$ is simple. Let $m(X)$ be the minimal polynomial for $\alpha$ then $k(\alpha) \cong k[X]/(m)$. From (3.36.39) $m(X)$ remains irreducible in $K[X]$ and therefore (using $k(\alpha) \otimes_k K \cong (k[X] \otimes_k K)/(m)$ as $K$-algebras, (3.4.42), and $k[X] \otimes_k K \cong K[X]$, (3.8.7))

$$K \otimes_k k(\alpha) \cong K[X]/(m)$$

is a field and in particular irreducible.

Conversely suppose that $K/k$ is geometrically irreducible and $\alpha \in K$ is separable. Then $k(\alpha)$ is geometrically irreducible by (3.36.32). Therefore, using $k(\alpha) \otimes_k \overline{k} \cong \left(k[X] \otimes_k \overline{k}\right)/(m_\alpha)$ as $\overline{k}$-algebras (3.4.42), then $k[X] \otimes_k \overline{k} \cong \overline{k}[X]$ (3.8.7), then by the Chinese Remainder Theorem (3.3.61)

$$k(\alpha) \otimes_k \overline{k} \cong k[X]/(m_\alpha) \otimes_k \overline{k} \cong \overline{k}[X]/(m_\alpha) \cong \overline{k}[X]/(X-\alpha_1)^{n_1} \times \cdots \times \overline{k}[X]/(X-\alpha_r)^{n_r}$$

is irreducible where $\alpha_1, ..., \alpha_r$ are the distinct roots in $\overline{k}$. This implies $r = 1$ and therefore $\alpha$ is purely inseparable (3.19.88). Being both separable and purely inseparable $\alpha \in k$ (3.19.89) as required. ∎

We made use of

**Lemma 3.36.39**
*Let $K/k$ be a field extension such that $k$ is algebraically closed in $K$. Then every irreducible polynomial $f \in k[X]$ remains irreducible in $K[X]$.*

*Proof.* Consider the extension $\overline{K}$ which contains $\overline{k}$. And suppose $g \mid f$ in $K[X]$. As $f$ splits completely in $\overline{k}$ then by unique factorisation in $\overline{K}[X]$ the coefficients of $g$ lie in $K \cap \overline{k} = k$. Therefore by irreducibility $g = f$ or is constant. In particular $f$ is irreducible in $K[X]$. ∎

**Proposition 3.36.40**
*Let $A$ be an integral $k$-algebra and $K := \mathrm{Frac}(A)$. Then $A$ is geometrically irreducible iff $K$ is geometrically irreducible.*

*Proof.* Suppose $K$ is geometrically irreducible and consider an arbitrary field extension $L/k$. Then we have an embedding $A \otimes_k L \hookrightarrow K \otimes_k L$, and therefore an embedding $(A \otimes_k L)_{\mathrm{red}} \hookrightarrow (K \otimes_k L)_{\mathrm{red}}$. By assumption these are integral and we conclude that $(A \otimes_k L)$ is irreducible.

Conversely suppose $A \otimes_k L$ is irreducible then define $\Omega := \mathrm{Frac}\left((A \otimes_k L)_{\mathrm{red}}\right)$. As $L \hookrightarrow (A \otimes_k L)$ is injective and $L$ is reduced we have a canonical map $L \hookrightarrow \Omega$. Similarly we have a canonical map $A \hookrightarrow \Omega$ and therefore a map $K \hookrightarrow \Omega$. Therefore we have a well-defined ring homomorphism

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$$\psi : K \otimes_k L \to \Omega$$
$$ab^{-1} \otimes \lambda \to (a \otimes \lambda) \cdot (b \otimes 1)^{-1}$$

We must show that if $x \in \ker(\psi)$ then $x$ is nilpotent. By definition

$$x = \sum_{i=1}^{n} \lambda_i \otimes \mu_i$$

We may find $a \in A$ such that $y = ax$ is of the form

$$y = \sum_{i=1}^{n} a_i \otimes \mu_i \in A \otimes_k L$$

We observe $y \in \ker(A \otimes_k L \to \Omega)$ whence $y$, and therefore $x$, is nilpotent. This means we have an embedding

$$(K \otimes_k L)_{\text{red}} \hookrightarrow \Omega$$

which means $(K \otimes_k L)_{\text{red}}$ is integral and therefore $K \otimes_k L$ is irreducible. ∎

**Proposition 3.36.41**
*Let $A$ be an integral $k$-algebra and $K \coloneqq \operatorname{Frac}(A)$. Then the following are equivalent*

*a) $A$ is geometrically irreducible*
*b) $A \otimes_k k'$ is irreducible for every finite separable extension $k'/k$*
*c) $A \otimes_k \overline{k}$ is irreducible*
*d) $K$ is geometrically irreducible*
*e) $K$ is relatively separably closed*

*Proof.* We may combine (3.36.36), (3.36.40) and (3.36.38). ∎

## 3.36.4 Geometrically Integral Algebras

Clearly a geometrically (resp. algebraically) integral algebra is integral, we are interested in providing sufficient conditions for the reverse implication.

**Proposition 3.36.42**
*Let $A$ be a geometrically integral $k$-algebra and $B$ an integral $k$-algebra. Then $A \otimes_k B$ is an integral domain.*

*Proof.* Let $L = \operatorname{Frac}(B)$ then $A \otimes_k B$ is a subring of $A \otimes_k L$ and we are done. ∎

**Proposition 3.36.43 (Criteria for Geometrically Integral in terms of Quotient Field)**
*Let $A$ be an integral $k$-algebra and $K \coloneqq \operatorname{Frac}(A)$. Then $A \otimes_k L$ is integral if and only if $K \otimes_k L$ is integral.*

*In particular $A$ is geometrically (resp. algebraically) integral if and only if $K$ is geometrically (resp. algebraically) integral.*

*Proof.* This follows exactly the same lines as (3.36.40). ∎

**Proposition 3.36.44 (Criteria to be Geometrically Integral)**
*Let $A$ be a $k$-algebra. Then the following are equivalent*

*a) $A$ is geometrically integral*
*b) $A$ is geometrically reduced and irreducible*
*c) $A \otimes_k \overline{k}$ is an integral domain*
*d) $A \otimes_k k'$ is an integral domain for every finite extension $k'/k$*

*Note we may restrict d) to only finite separable, and finite purely inseparable extensions of height at most 1.*

*Proof.* This is largely a collection of existing results. $a) \iff b)$ is by definition and (3.3.66). Then $b) \iff c) \iff d)$ is from the reduced (3.36.26) and irreducible (3.36.36) cases, together with the trivial observation that reduced + irreducible $\iff$ integral (3.3.66). ∎

**Proposition 3.36.45 (Regular Extension)**
*Let $K/k$ be a finitely generated field extension. Then the following are equivalent*

*a) $K/k$ is geometrically integral*
*b) $K/k$ is separably generated and relatively separably closed*
*c) $K/k$ is separably generated and relatively algebraically closed*

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*d)* $K \otimes_k k'$ *is an integral domain for every* finite *extension* $k'/k$
*e)* $K \otimes_k k'$ *is a field for every* algebraic *extension* $k'/k$
*f)* $K \otimes_k \overline{k}$ *is an integral domain*
*g)* $K \otimes_k \overline{k}$ *is a field*

*We say such an extension is* ***regular****.*

*Proof.* *a)* $\implies$ *b)* is (3.36.29) and (3.36.38). *c)* $\implies$ *b)* is immediate. For the converse suppose $\alpha \in K/k$ is algebraic then $k(\alpha)/k$ is geometrically reduced (3.36.29) and therefore separable algebraic (3.36.18). By hypothesis $\alpha \in k$ as required, giving *b)* $\iff$ *c)*. For *b)* $\implies$ *a)*: $K/k$ being separably generated, it is geometrically reduced (3.36.24), and being relatively separably closed, it is geometrically irreducible (3.36.38) together these give *a)* (3.3.66). This closes the loop *a)* $\iff$ *b)* $\iff$ *c)*. *a)* $\iff$ *d)* $\iff$ *f)* was already proven in (3.36.44), and *f)* $\iff$ *g)* was proven in (3.35.57). *e)* $\implies$ *d)*, *g)* is immediate.

For *f)* $\implies$ *e)* fix an algebraic $k'/k$ and an embedding $k' \hookrightarrow \overline{k}$ (exists by (3.19.61)); the canonical map $K \otimes_k k' \to K \otimes_k \overline{k}$ is injective (3.36.2) ($k'$ being a $k$-subalgebra of $\overline{k}$), so $K \otimes_k k'$ is a subring of the domain $K \otimes_k \overline{k}$, hence itself a domain; as $k'/k$ is algebraic, this domain is moreover a field (3.35.57). ∎

**Corollary 3.36.46**
*Let $A$ be a finitely-generated integral $k$-algebra and $K$ the quotient field. Then the following are equivalent*

*a)* *$A$ is geometrically integral*
*b)* *$K/k$ is geometrically integral*
*c)* *$K/k$ is separably generated and relatively separably closed*
*d)* *$K \otimes_k k'$ is an integral domain for every finite extension $k'/k$*
*e)* *$K \otimes_k \overline{k}$ is an integral domain*
*f)* *$K \otimes_k \overline{k}$ is a field*

*Proof.* *a)* $\iff$ *b)* is (3.36.43) and the rest is (3.36.45). ∎

The mechanism underlying (6.3.44) below (base-changing a place of an algebraic function field along an algebraic extension of the constant field) is really a fact about discrete valuation rings, unrelated to the transcendence degree 1 setting of function fields; we isolate it here.

**Proposition 3.36.47 (Unramified Base Extension of a Discrete Valuation Ring)**
*Let $k$ be a perfect field and $\mathcal{O}$ a discrete valuation ring containing $k$, with fraction field $F$ and maximal ideal $\mathfrak{m}$, such that $F/k$ is geometrically integral and $k(\mathfrak{m}) := \mathcal{O}/\mathfrak{m}$ is finite over $k$.*

*Let $k'/k$ be algebraic, write $\mathcal{O}_{k'} := \mathcal{O} \otimes_k k'$ and $\mathfrak{b} := \mathfrak{m}\mathcal{O}_{k'}$. Then*

*a)* *$\mathcal{O}_{k'}$ has exactly $r$ maximal ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_r$, the preimages of those of $\mathcal{O}_{k'}/\mathfrak{b}$*
*b)* *the localisation $(\mathcal{O}_{k'})_{\mathfrak{p}_j}$ is a place of $F_{k'}$ unramified over $\mathcal{O}$: a uniformiser of $\mathcal{O}$ generates the maximal ideal $\mathfrak{p}_j(\mathcal{O}_{k'})_{\mathfrak{p}_j}$, and*
*c)* *every valuation ring $R$ of $F_{k'}$ with $k' \subset R$, $\mathcal{O} \subset R$, and $R \cap F = \mathcal{O}$ is of the form $R = (\mathcal{O}_{k'})_{\mathfrak{p}_j}$ for some $j$, and*
*d)* *when $k'/k$ is finite then $(\mathcal{O}_{k'})_{\mathfrak{p}_j}$ is a discrete valuation ring, and*
*e)* *$(\mathcal{O}_{k'})_{\mathfrak{p}_j} \cap F = \mathcal{O}$ and $\mathfrak{p}_j(\mathcal{O}_{k'})_{\mathfrak{p}_j} \cap F = \mathfrak{m}$, inside $F_{k'}$: that is $(\mathcal{O}_{k'})_{\mathfrak{p}_j}$ dominates $\mathcal{O}$ and restricts to it. Together with c), the valuation rings $R$ of $F_{k'}$ with $k' \subset R$, $\mathcal{O} \subset R$ and $R \cap F = \mathcal{O}$ are therefore exactly the $(\mathcal{O}_{k'})_{\mathfrak{p}_j}$.*

*Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.*

Let $\pi$ be a uniformiser of $\mathcal{O}$, so $\mathfrak{m} = (\pi)$ (3.31.2). Since $F/k$ is geometrically integral and $k'/k$ algebraic, $F_{k'}$ is a field (3.36.45), and $\mathcal{O}_{k'}$ is a subring (...) and in particular an integral domain. Further $\mathcal{O}_{k'}$ integral over $\mathcal{O}$ (3.24.20).

$\mathcal{O}_{k'}/\mathfrak{b} = \mathcal{O}_{k'}/\mathfrak{m}\mathcal{O}_{k'} \cong (\mathcal{O}/\mathfrak{m}) \otimes_k k' = k(\mathfrak{m}) \otimes_k k'$ as $k$-algebras (3.4.42): a finite-dimensional (as $k(\mathfrak{m})/k$ is finite, by hypothesis) reduced ($k(\mathfrak{m})/k$ and $k'/k$ both separable, $k$ being perfect, (3.19.103), (3.36.28)) $k$-algebra, hence a finite product of $r$ fields (3.36.12), with every prime maximal (3.36.13).

**a) Finitely many maximal ideals.** $\mathcal{O}_{k'}/\mathfrak{b}$, being a finite product of $r$ fields (above), has finitely many maximal ideals $\mathfrak{p}'_1, \dots, \mathfrak{p}'_r$, all its primes maximal (3.36.13). Every maximal ideal of $\mathcal{O}_{k'}$ contracts to a maximal ideal of $\mathcal{O}$ (3.24.16), that is to $\mathfrak{m}$ ($\mathcal{O}$ being local), hence contains $\mathfrak{b}$; so, by the ideal correspondence for $\mathcal{O}_{k'} \twoheadrightarrow \mathcal{O}_{k'}/\mathfrak{b}$ (3.3.55), the maximal ideals of $\mathcal{O}_{k'}$ are exactly the preimages $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ of $\mathfrak{p}'_1, \dots, \mathfrak{p}'_r$ – and, every prime of the quotient being maximal, these are also *every* prime of $\mathcal{O}_{k'}$ containing $\mathfrak{b}$.

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As $\mathfrak{b}$ is radical (the quotient being reduced), the Prime Nullstellensatz (3.3.45) gives $\mathfrak{b} = \mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_r$; as the $\mathfrak{p}_j$ are pairwise comaximal (distinct maximal ideals), this equals $\mathfrak{p}_1 \cdots \mathfrak{p}_r$ (3.3.61).

**b) Unramifiedness.** Write $\mathcal{O}_j := (\mathcal{O}_{k'})_{\mathfrak{p}_j}$. By (3.6.18).a) and multiplicativity of localisation on ideals (3.30.8),

$$\pi\mathcal{O}_j = \mathfrak{b}\mathcal{O}_j = \mathfrak{b}_{\mathfrak{p}_j} = (\mathfrak{p}_1)_{\mathfrak{p}_j} \cdots (\mathfrak{p}_r)_{\mathfrak{p}_j} = \mathfrak{p}_j\mathcal{O}_j$$

(the first equality as $\mathfrak{b} = \pi\mathcal{O}_{k'}$; the factors $i \neq j$ becoming the unit ideal, as $\mathfrak{p}_i \not\subset \mathfrak{p}_j$). So $\pi$ generates $\mathfrak{p}_j\mathcal{O}_j$, once $\mathcal{O}_j$ is known to be a valuation ring of $F_{k'}$ properly between $k'$ and $F_{k'}$ – established next.

**d) The finite case: $\mathcal{O}_j$ is a DVR.** Suppose $k'/k$ is finite. Then $\mathcal{O}_{k'}$ is a finite, free $\mathcal{O}$-module (3.4.26), hence a Noetherian $\mathcal{O}$-module (3.15.4) ($\mathcal{O}$ Noetherian, being a PID (3.16.2)), so Noetherian as a ring (3.15.3) (every ideal being in particular an $\mathcal{O}$-submodule). So $\mathcal{O}_j$ is Noetherian (3.15.7), local with maximal ideal $\mathfrak{p}_j\mathcal{O}_j$ (3.6.42), and a domain (above). It has Krull dimension 1: as $\mathcal{O} \hookrightarrow \mathcal{O}_{k'}$ is injective and integral (module-finite), $\dim \mathcal{O}_{k'} = \dim \mathcal{O} = 1$ (3.24.22) ($\mathcal{O}$ a DVR, hence of dimension 1 (3.31.2)); as $\mathcal{O}_{k'}$ is a domain, $(0) \subsetneq \mathfrak{p}_j$ gives $\operatorname{ht}(\mathfrak{p}_j) \geq 1$, while $\operatorname{ht}(\mathfrak{p}_j) \leq \dim \mathcal{O}_{k'} = 1$, so $\dim \mathcal{O}_j = \operatorname{ht}(\mathfrak{p}_j) = 1$ (3.22.4).b). With $\mathfrak{p}_j\mathcal{O}_j = (\pi)$ (above), $\mathcal{O}_j$ is a DVR (3.31.2).

**In general, $\mathcal{O}_j$ is a valuation ring.** Write $k' = \bigcup_i k'_i$ as the directed union of its finite subextensions (3.19.47), so $\mathcal{O}_{k'} = \bigcup_i \mathcal{O}_{k'_i}$ for $\mathcal{O}_{k'_i} := \mathcal{O} \otimes_k k'_i$ (a directed union of subrings, via (3.36.2)). For each $i$, $\mathfrak{p}_{i,j} := \mathfrak{p}_j \cap \mathcal{O}_{k'_i}$ contains $\mathfrak{m}\mathcal{O}_{k'_i}$ (as $\mathfrak{p}_j \supseteq \mathfrak{b}$), so corresponds, under the ideal correspondence for the finite-dimensional ring $\mathcal{O}_{k'_i}/\mathfrak{m}\mathcal{O}_{k'_i} = k(\mathfrak{m}) \otimes_k k'_i$ (as $k$-algebras, (3.4.42)) (reduced, as $k(\mathfrak{m})/k$ and $k'_i/k$ are separable, $k$ perfect (3.19.103), (3.36.28)), so a finite product of fields (3.36.12), as in the statement, applied to $k'_i$ in place of $k'$), to a prime of that ring, automatically maximal (3.36.13) so $\mathfrak{p}_{i,j}$ is a maximal ideal of $\mathcal{O}_{k'_i}$. By the finite case just established (taking $k'_i$ for $k'$), $\left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$ is a DVR of $F_{k'_i}$, with $\mathcal{O} \subsetneq \left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$ and $\pi$ generating $\mathfrak{p}_{i,j}\left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$.

**Claim: $\mathcal{O}_j = \bigcup_i \left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$.** For $k'_i \subseteq k'_l$, $\mathfrak{p}_{l,j} \cap \mathcal{O}_{k'_i} = \mathfrak{p}_j \cap \mathcal{O}_{k'_i} = \mathfrak{p}_{i,j}$, so (3.6.9) gives $\left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}} \subseteq \left(\mathcal{O}_{k'_l}\right)_{\mathfrak{p}_{l,j}}$: a directed system. As $\mathcal{O}_{k'} = \bigcup_i \mathcal{O}_{k'_i}$ is a directed union of subrings with $\mathfrak{p}_{i,j} = \mathfrak{p}_j \cap \mathcal{O}_{k'_i}$, (3.6.10) gives $\mathcal{O}_j = \bigcup_i \left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$.

**Claim: $k' \subsetneq \mathcal{O}_j \subsetneq F_{k'}$.** As $\pi$ generates the maximal ideal of every $\left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$, (3.31.6) applies directly (with $F_{k'} = \bigcup_i F_{k'_i}$ the corresponding directed union of fields): $\mathcal{O}_j = \bigcup_i \left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$ is a valuation ring of $F_{k'}$ with maximal ideal $(\pi)\mathcal{O}_j$, and $\mathcal{O}_j \subsetneq F_{k'}$. Also $k' \subsetneq \mathcal{O}_j$: $k' = \bigcup_i k'_i \subseteq \bigcup_i \left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}} = \mathcal{O}_j$ is immediate, while $\pi \in \mathcal{O}_j \setminus k'$, as $\pi \notin k'_i$ for every $i$ (else $\pi \in \mathfrak{p}_{i,j} \cap k'_i$ would be a unit of $\mathcal{O}_{k'_i}$ (nonzero elements of the subfield $k'_i \subset \mathcal{O}_{k'_i}$ are units), contradicting $\pi \notin \left(\left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}\right)^\times$), so $\pi \notin k'$.

This closes b): $\mathcal{O}_j$ is a valuation ring of $F_{k'}$ with $k' \subsetneq \mathcal{O}_j \subsetneq F_{k'}$ – that is, a place of $F_{k'}$ – and $\pi$ generates both $\mathfrak{m}$ and $\mathfrak{p}_j\mathcal{O}_j$ (above): the same uniformiser generating the maximal ideal on both sides of $\mathcal{O} \subset \mathcal{O}_j$ is exactly what it means for $\mathfrak{m}$ to be unramified in $\mathcal{O}_{k'}$ – matching the factorisation $\mathfrak{b} = \mathfrak{p}_1 \cdots \mathfrak{p}_r$ above. When $k'/k$ is finite, $\mathcal{O}_j$ is moreover a discrete valuation ring (d)), recovering the usual ramification index $e(\mathcal{O}_j/\mathcal{O}) = v_{\mathcal{O}_j}(\pi) = 1$.

**c) Every valuation ring $R$ with $R \cap F = \mathcal{O}$ is some $\mathcal{O}_j$.** Let $R$ be a valuation ring of $F_{k'}$ with $k' \subset R$, $\mathcal{O} \subset R$, and $R \cap F = \mathcal{O}$. As $\mathcal{O}_{k'}$ is generated as a ring by (the images of) $\mathcal{O}$ and $k'$ (spanned by simple tensors $a \otimes b = a \cdot b$), both contained in $R$, $\mathcal{O}_{k'} \subset R$. So $\mathfrak{q} := \mathfrak{m}_R \cap \mathcal{O}_{k'}$ is prime (a contraction of the prime $\mathfrak{m}_R$ along $\mathcal{O}_{k'} \hookrightarrow R$); it contains $\mathfrak{b}$, as $\pi \in \mathfrak{m}_R$ (else $\pi^{-1} \in R$, so $\pi^{-1} \in R \cap F = \mathcal{O}$, contradicting $\pi$ a non-unit of $\mathcal{O}$), giving $\mathfrak{b} = \pi\mathcal{O}_{k'} \subset \mathfrak{m}_R \cap \mathcal{O}_{k'} = \mathfrak{q}$. So $\mathfrak{q}$ corresponds to a prime, automatically maximal (3.36.13), of the finite-dimensional ring $\mathcal{O}_{k'}/\mathfrak{b} \cong \prod_j E_j$; so $\mathfrak{q} = \mathfrak{p}_j$ for some $j$.

Every $s \in \mathcal{O}_{k'} \setminus \mathfrak{p}_j$ then satisfies $s \notin \mathfrak{q} = \mathfrak{m}_R \cap \mathcal{O}_{k'}$, that is $s \notin \mathfrak{m}_R$, so $s$ is a unit of $R$; by the universal property of localisation, $\mathcal{O}_j$ then embeds in $R$. As $\mathfrak{m}_{\mathcal{O}_j} = \pi\mathcal{O}_j$ and $\pi \in \mathfrak{m}_R$, $\mathfrak{m}_{\mathcal{O}_j} \subseteq \mathfrak{m}_R \cap \mathcal{O}_j$; conversely $\mathfrak{m}_R \cap \mathcal{O}_j$ is a proper ideal of the local ring $\mathcal{O}_j$, hence $\subseteq \mathfrak{m}_{\mathcal{O}_j}$: so $\mathfrak{m}_{\mathcal{O}_j} = \mathfrak{m}_R \cap \mathcal{O}_j$, that is $R$ dominates $\mathcal{O}_j$ (3.25.4). As $\mathcal{O}_j$ is a valuation ring, it is maximal under domination (3.25.13), so $R = \mathcal{O}_j$.

**e) $\mathcal{O}_j$ restricts to $\mathcal{O}$.** As $\mathcal{O}_{k'} \subset F_{k'}$ is a domain, $\mathcal{O}_j$ is the subring of $F_{k'}$ of fractions $a/s$ with $a \in \mathcal{O}_{k'}$, $s \in \mathcal{O}_{k'} \setminus \mathfrak{p}_j$, so $\mathcal{O} \subset \mathcal{O}_{k'} \subset \mathcal{O}_j$. Moreover $\mathfrak{m} \subseteq \mathfrak{m}\mathcal{O}_{k'} = \mathfrak{b} \subseteq \mathfrak{p}_j$ (a)), so $\mathfrak{m} \subseteq \mathfrak{p}_j\mathcal{O}_j = \mathfrak{m}_{\mathcal{O}_j}$ (3.6.42): and $\mathfrak{p}_j\mathcal{O}_j$ is a proper ideal. Since $\mathcal{O}$ is a valuation ring of $F$, (3.25.6) (with $B = \mathcal{O}_j$, $\mathfrak{n} = \mathfrak{p}_j\mathcal{O}_j$) gives $\mathcal{O}_j \cap F = \mathcal{O}$ and $\mathfrak{p}_j\mathcal{O}_j \cap F = \mathfrak{m}$. Combined with c) this shows that the valuation rings $R$ of $F_{k'}$ with $k' \subset R$, $\mathcal{O} \subset R$ and $R \cap F = \mathcal{O}$ are precisely $\mathcal{O}_1, \ldots, \mathcal{O}_r$. ∎

**Remark 3.36.48 (Relation to the Stacks Project)**
*For $k'/k$ finite, this result also follows by combining two general facts from [Stacks]: unramified ring maps are stable under base change ([Stacks] (Tag 00US), applied to the finite separable, hence unramified, extension $k \to k(\mathfrak{m})$, base-*

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*changed along $k \to k'$), and a finite-dimensional algebra over a field splits, after any base change, into a product of local factors dual to its maximal ideals (the Stacks Project analogue of (3.36.15), (3.36.13) above).*

*For $k'/k$ merely* algebraic*, the Stacks Project's treatment of unramified morphisms is developed for morphisms of finite type, so does not directly cover this case; the passage to infinite $k'$ via the directed union of finite subextensions, as in* In general, $\mathcal{O}_j$ is a valuation ring *above, has no immediate analogue there and is instead the kind of limit argument found in the Stacks Project's chapter on limits of rings (e.g. [Stacks] (Tag 07RN ff.), on writing a ring as a filtered colimit and transporting properties of the terms to the colimit).*

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